Digital Surface Model-Embedded Intrinsic Hyperspectral Unmixing

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***Abstract*— High-precision spectral unmixing (SU) of hyperspectral image (HSI) faces a challenging problem in that it is difficult to distinguish the different objective classes presenting similar spectra without elevation information. A digital surface model (DSM) with the same spatial resolution as HSI can provide additional geometric information that can be useful for the HSI SU task. Existing SU methods that incorporate the DSM data miss the consideration that DSM data and hyperspectral data are actually different dimensions of information from different sensors observing the same scene. The intrinsic hyperspectral unmixing model can obtain the shading component and the reflectance component, which can be further decomposed into the endmember and the abundance. From a physical modeling perspective, the shadow component can be considered as the interaction of illumination and the geometric component from the DSM data; thus, the quality of hyperspectral unmixing under invariant illumination conditions can be significantly enhanced. Furthermore, the elevation information within the DSM data contributes to the unmixing process by distinguishing different objective classes with similar physiochemical properties at varying altitudes. Experimental validation is conducted using three HSI datasets. The results can indicate the robustness and superiority of the proposed unmixing method.**

***Index Terms*— Digital surface model (DSM), intrinsic hyperspectral unmixing, physical imaging process, spectral variability.**

NOMENCLATURE

Symbol	Meaning
N	Number of pixels.
L	Number of bands.
P	Number of endmembers.
$\boldsymbol{y}_n$	Spectral vector with n pixels.
$\mathbf{E}_0$	Endmember matrix.
$\mathbf{A}$	Abundance matrix.
$\mathbf{S}$	Shading component.
$\mathbf{R}$	Reflectance component.
$\mathbf{I}$	Direction of the incident light.
$\mathbf{n}$	Surface normal.
$\mathbf{G}$	Geometric component.
$\mathbf{K}$	Elevation.
$\mathbf{M}$	Illumination parameters.

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I. INTRODUCTION

THE rapid development of hyperspectral imaging technology has enabled its widespread use [1]. Spectral unmixing (SU), as a unique downstream task in the hyperspectral field, is defined as the decomposition of the mixed pixels in a hyperspectral image (HSI) into a set of endmembers that characterize the feature properties and the corresponding abundance [5], which is also widely used in many areas [6], [7].

In order to estimate the endmembers and the corresponding abundances accurately, various SU methods have been developed [8], which can be classified as linear mixing model (LMM) and nonlinear mixing model (NLMM). LMM takes a macroscopic viewpoint and assumes that light is picked up by the sensor after direct interaction with the surface of the feature without interaction between features. The NLMM model considers the interaction between features at a microlevel, making it more realistic. Due to the complexity of the NLMM and the number of parameters involved, it is often difficult to generalize the model to different scenarios. In recent years, deep learning research has been growing rapidly [15], [16], [17], [18], [19]. The unmixing algorithms of deep learning can extract features from a data-driven perspective and can be optimized through network learning [20]. The untied denoising autoencoder (AE) with sparsity (uDAS) [20] offers an effective improvement in SU by introducing a decoupled denoising AE and sparse features. Ozkan et al. [21] leverage a sparse AE network structure to achieve excellent endmember extraction capabilities. Gao et al. [24] achieve hyperspectral nonlinear unmixing by utilizing nonnegative tensor factorization and low-rank representation of abundance maps, thereby reducing data complexity and minimizing the influence of noise while accurately capturing nonlinear features in the data. However, these methods may require significant computational resources and time when handling large-scale hyperspectral data, and their generalization ability needs further validation. Most existing SU algorithms are based on the assumption that the spectral curves of the target remain constant, which ignores the phenomenon of spectral variability caused by changing imaging conditions such as atmosphere and illumination [26], [27]. As a result, many unmixing algorithms for spectral variability have received widespread and sustained consideration [28]. These algorithms can improve unmixing results by incorporating scaling or additive factors to model spectral variability. Thouvenin et al. [29] and Drumetz et al. [30] model spectral variability primarily by introducing

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additive factors and scale factors along with the nonlinear factors, respectively. Yang [34] employs a nonlinear unmixing algorithm for shadow compensation, enabling the extraction of shadow-free components from images under varying lighting conditions, thus enhancing the overall unmixing quality. Zhang et al. [35] integrate nonlinear components with the traditional LMM, allowing for the modeling of more complex imaging conditions. In addition, by partitioning the shadowed and nonshadowed regions, the method effectively mitigates the influence of shadows on the unmixing results. Furthermore, targets with analogous physicochemical characteristics can display highly similar spectral curves in remote sensing images. Extracting features from a single HSI for SU may result in missing structures and inaccurate results. It is important to consider using multiple data for more accurate SU. Methods for the fusion of spatial features to improve the accuracy of SU have received increasing attention in the last few years [36], [37]. In recent years, there has been a growing interest in using more efficient spectral and spatial techniques through multimodal remote sensing data [38]. Different imaging sensors on multiple platforms can acquire many different types of remote sensing image data, such as active devices and passive devices, including light detection and ranging (LiDAR), multispectral image (MSI), and HSI. LiDAR data are 3-D coordinate data recorded after scanning a 2-D space, which has the characteristics of high density, high accuracy, and fast generation efficiency, and can provide meaningful spatial information for the scene. LiDAR is now widely used in topographic mapping of various architectural, urban geological, and hydrological environments [47]. LiDAR point cloud data can be used to generate digital surface models (DSMs), which provide surface elevation data for features such as land surfaces and buildings. The spatial resolution of sampled DSM data aligns with that of hyperspectral imagery, with a single band representing elevation values for each pixel. In real-world scenarios, differences in elevation are critical to the distinction between different objects. It is worth noting that along with the rapid advancement of sensors, LiDAR data and hyperspectral data can be acquired simultaneously during the imaging process [50]. The synergistic use of these two types of data also has great potential. Despite the many advances in LiDAR point cloud data technology for topographic mapping, the use of LiDAR data for SU is still in its infancy. Considering the incorporation of such data into spatial regularization, feature enhancement can be effectively achieved. However, existing SU methods that use multimodal hyperspectral data always focus on spatial feature enhancement rather than considering the imaging process of the multimodal data with the scene.

Since a DSM acquired from LiDAR data can capture surface features with high spatial resolution, which are not significantly influenced by external factors, a novel DSM data-embedded intrinsic hyperspectral unmixing model is proposed. In this model, DSM data can infer scene properties such as illumination and the surface normal. Specifically, we decompose the original HSI data into the reflectance component and the shading component [51], while the reflectance component can be used to unmix and the shading component can be considered as the interaction of illumination and surface normal from the DSM data.

The contributions and the main innovations of this article are as follows.

1) A DSM-embedded multimodal hyperspectral unmixing model is proposed, where the elevation information derived from DSM data serves as both a source of data for the physical imaging model and a regularization term for prior constraints, enabling effective unmixing of objective classes with similar spectral signatures.
2) From the perspective of physical imaging, the process of spatial information-driven decomposition of the shading component in DSM is elucidated, thereby facilitating the hyperspectral unmixing process under conditions of invariant illumination.

The rest of this article has the following structure. Section II is first an introduction to the related work. Section III is then devoted to the description of the proposed model. In Section IV, the experimental results are presented with the corresponding discussions. Finally, conclusions are drawn in Section V.

II. Related Work

In this section, we analyze some of the existing typical multimodal hyperspectral unmixing algorithms in terms of both model-driven methods and data-driven methods [52], and then, the fundamental imaging theory of intrinsic image decomposition (IID) [46] is introduced.

A. Model-Driven Model

The classical LMM has been widely used to decompose a mixed pixel into a collection of endmembers and abundances. LMM represents a mixed spectral vector combined by the mixed pixel as the linear combination of the endmember and the abundance

$$y_n = \mathbf{E_0} a_n + n_n \tag{1}$$

where $\boldsymbol{y}_n, n = 1, \ldots, N \in \mathbb{R}^{L\times 1}$ represents the spectral vector, L is the band of HSI, N is the number of the pixels, $\mathbf{E}_0 = [e_{0,1}, \ldots, e_{0,p}]$ is an endmember matrix, P represents the number of endmembers, and a_n is the abundance vector.

The above equation can be expressed as

$$\mathbf{Y} = \mathbf{EA} + \mathbf{N} \tag{2}$$

where $\mathbf{Y} \in \mathbb{R}^{L\times N}$ is the satellite his; $\mathbf{E} \in \mathbb{R}^{L\times P}$ is the endmember matrix; $\mathbf{A} \in \mathbb{R}^{P\times N}$ is the abundance matrix; and $\mathbf{N} \in \mathbb{R}^{L\times N}$ is an addictive term, including the modeling error and the observed noise. The abundance matrix is subject to two priori constraints: the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC).

The optimization can be summarized as

$$\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{Y} - \mathbf{EA}\|_F^2 + \lambda\phi(\mathbf{A}) \tag{3}$$

where λ is the penalty parameter to balance the modeling term and the spatial regularization.

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In a single HSI, spatial regularization is always characterized as l_p norms

$$\phi(\mathbf{A}) = \sum_{i=1}^{N} \sum_{j \in \varsigma(i)} \|a_i - a_j\|_p^p \tag{4}$$

where $\varsigma(i)$ is the collection of pixels in the neighborhood of the pixel i. In particular, the norm p is often set to 1 or 2 to facilitate the accuracy of SU.

The quality of the related work has been improved by introducing the weight of spatial regularization

$$\phi(\mathbf{A}) = \sum_{i=1}^{N} \sum_{j \in \varsigma(i)} w_{ij} \|a_i - a_j\|_p^p \tag{5}$$

where the weight w_{ij} is introduced to control the spatial regularization of the neighboring pixels associated with different objects differently.

Since multimodal SU is still in the beginning stage, Uezato et al. [39] are more representative, as they incorporate external LiDAR to adjust the standard spatial regularization during the unmixing process. Specifically, Uezato et al. [39] introduce a new weight from DSM data by calculating the similarity between adjacent pixels based on their heights as

$$w_{ij} = \frac{1}{\mathbb{Q}_i} \exp\left(-\frac{1}{\sigma_h^2} \frac{(h_i - h_j)^2}{(h_i + h_j)^2}\right) \tag{6}$$

where h_i and h_j are the heights of pixel i and pixel j from DSM data, respectively. Multimodal SU can be effectively achieved using the spatial elevation information to weight the spatial regularization accordingly. Still, spatial regularization mainly focuses on the shallow feature rather than the interaction of multimodal data currently.

B. Data-Driven Model

As a representative of the deep learning methods for SU, the AE, which includes an encoder and a decoder, can learn low-dimensional representation and the corresponding weight base simultaneously by minimizing the reconstruction error [53].

Generally, the encoder part can be expressed as

$$x_i = f_{\text{en}}(y_i) = f(\mathbf{W}^T y_i + b) \tag{7}$$

where $f()$ is an activation function; and $\mathbf{W}$ and b are the weight and the bias, respectively.

The decoder part can be expressed as

$$\tilde{y}_i = f_{\text{de}}(x_i) = \tilde{\mathbf{W}}^T x_i \tag{8}$$

where $\tilde{\mathbf{W}}$ is the decoder weight and $\tilde{y}_i$ is the reconstructed output.

As can be seen, these AE-based approaches focus on a single HSI and fail to discriminate different but similar objects from the same material in the real scene.

Typically, a novel multimodal hyperspectral unmixing network [24] was proposed using LiDAR-generated attention maps to focus on spatial information. Specifically, to better model the spatial information of LiDAR, an attribute profile (AP) is used to extract the geometrical structure of different objects. The design of the network starts since LiDAR can be highly informative to differentiate between spectrally similar substances. More spatial information can be fused in SU so that desirable unmixing results can be obtained. The inclusion of squeeze-and-excitation (SE) attention allows the effective fusion of HSI and LiDAR features in an unsupervised way. Nowadays, spatial regularization mainly focuses on the shallow feature rather than the interaction of multimodal data currently. Still, the problem of spectral variability remains a challenge in multimodal hyperspectral unmixing.

C. Multimodal IID

The IID method can decompose the HSI into a shading component, which reflects the lighting of the original image, and the reflectance component, which refers to the portion of the image that can remain unchanged under changing light conditions.

For HSI $\mathbf{Y} \in \mathbb{R}^{L \times N}$, the IID can be expressed as

$$\mathbf{Y} = \mathbf{S} \odot \mathbf{R} \tag{9}$$

where $\mathbf{S} \in \mathbb{R}^{L \times N}$ is the shading component, which can characterize the interaction of illumination with the surface normal of the material. $\mathbf{R} \in \mathbb{R}^{L \times N}$ is the reflectance component, which is only related to the inherent characteristics of the materials themselves. In this way, $\mathbf{R}$ can always characterize the spectra of materials accurately, reducing spectral uncertainty resulting from changes in external imaging conditions. For a specified type of material, its reflectance component $\mathbf{R}$ should remain consistent, with values ranging between 0 and 1. $\odot$ is the Schur–Hadamard product.

The spatial information can be introduced in multimodal hyperspectral unmixing in this way

$$\mathbf{S} = \langle \mathbf{n}, \mathbf{I} \rangle \tag{10}$$

where $\mathbf{n} = [N_x, N_y, N_z]^T$ is the surface normal and $\mathbf{I} = [\mathbf{I}_x, \mathbf{I}_y, \mathbf{I}_z]^T$ is the direction of the incident light. Considering that spherical-harmonic (SH) lighting is a lighting algorithm based on SHs, the shading image $\mathbf{S}$ can also be considered as the interaction of the geometric from the DSM data with the illumination to subsequently be able to better optimize the solution [45]

$$\mathbf{SH}(\mathbf{n}, \mathbf{I}) = \mathbf{GM}\mathbf{1}_N^T \tag{11}$$

where $\mathbf{G}$ is defined as the geometric component, $\mathbf{G} = [\mathbf{G}_1, \mathbf{G}_2, \ldots, \mathbf{G}_n]^T$, in which $\mathbf{G}_n = [\mathbf{G}_n^x, \mathbf{G}_n^y, \mathbf{G}_n^z]^T$. $\mathbf{G}$ is the set of geometric components of pixels, where n represents the number of hyperspectral pixels and $\mathbf{G}_n$ denotes the geometric component value of the nth pixel. $\mathbf{G}$ can be computed by the elevation matrix $\mathbf{K}$ from the DSM data, where $\mathbf{K}$ represents the elevation of the hyperspectral with n pixels, $\mathbf{K} = [\mathbf{K}_1, \mathbf{K}_2, \ldots, \mathbf{K}_n]^T$. $\mathbf{M}$ is defined as the first- and second-order spherical-harmonic illumination parameters, $\mathrm{M} = [\mathbf{M}_1, \mathbf{M}_2, \ldots, \mathbf{M}_9]$. The priori constraint on illumination $\mathbf{M}$ here is a multivariate Gaussian fit to SH illumination. The advantage of SH illumination is that the amount of computation at runtime is independent of the number of light

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sources, and if the parameters are sufficient, it is possible to simulate real-time illumination results better.

Generally, the regularization constraint matrix can be summarized as smoothing prior, which means that there is a greater probability that pixels in neighboring positions have the same or similar reflectance

$$\mathbf{C}_i = w_{ij}\mathbf{C}_j \tag{12}$$

where $\mathbf{C}_i$ and $\mathbf{C}_j$ are the reflectance of the pixel i and the pixel j, respectively; and w_{ij} is the spectral similarity. However, this prior cannot be satisfied at the boundary of the different targets. Based on this, a sparse constraint matrix $\mathbf{W}$ can be used to better represent the solution of the reflectance image $\mathbf{R}$

$$C_1(\mathbf{R}) = \frac{1}{2}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_F^2 \tag{13}$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm.

III. Proposed Approach

A. Motivation

Combining (2)–(9), the adjustment key function of intrinsic SU can be formulated as follows [28]:

$$\mathbf{R} = \mathbf{E}\mathbf{A} + \mathbf{N}. \tag{14}$$

Considering that the spatial information from the DSM data can model the shading component more accurately from the process of physical imaging.

Based on this, the DSM-embedded intrinsic unmixing model can be formulated as

$$\mathbf{Y} = \mathbf{S} \odot \mathbf{R} = \left(\mathbf{G}\mathbf{M}\mathbf{1}_N^T\right) \odot (\mathbf{E}\mathbf{A}). \tag{15}$$

It is clear that the observed HSI $\mathbf{Y}$ is the interplay of the shading component $\mathbf{S}$ and the reflectance component $\mathbf{R}$, while $\mathbf{S}$ can be decomposed into the geometric information from the DSM data and illumination, and $\mathbf{R}$ can be used to process the SU. The framework of the proposed Digital surface model-embedded intrinsic hyperspectral unmixing algorithm is illustrated in Fig. 1.

The above model is a typical ill-posed problem and the key to this problem is to design accurate constraint terms about the abovementioned variables.

B. Proposed Model

To solve the abovementioned problem, we can construct the following objective function based on the model supplied:

$$\begin{aligned} U = &\frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\alpha}{2}\left\|\mathbf{S} - \mathbf{G}\mathbf{M}\mathbf{1}_N^T\right\|_F^2 + \frac{\beta}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 \\ &+ \Psi(\mathbf{R}) + \Upsilon(\mathbf{A}) \\ \text{s.t. } &\mathbf{A} \geq 0 \end{aligned} \tag{16}$$

where the physical meaning of the first term is keeping the residual error between the original HSI and the reconstructed HSI as small as possible, the second term is to minimize the residual error between the shading image and the interplay of the geometric component from the DSM data and the illumination component, the third term is to minimize the residual error between the reflectance image and the reconstructed reflectance during the process of unmixing. The last two terms $\Psi(\mathbf{R})$ and $\Upsilon(\mathbf{A})$ represent the reflectance constraints and abundance constraints, respectively.

Here, we introduce the reflectance priors and the abundance prior to simplifying the abovementioned problem.

Reflectance Prior 1: It is more likely that the spectra will be similar for targets in neighboring pixels. Thus, the reflectance of pixel i can be expressed as a linear combination of the reflectance of its neighboring pixels

$$R_i = w_{ij}R_j \tag{17}$$

where R_i and R_j denote the reflectance of the pixel i and the pixel j, respectively; and w_{ij} is the spectral similarity between the pixel i and the pixel j.

The matrix form can be given as

$$\mathbf{R} = \mathbf{W}\mathbf{R}. \tag{18}$$

Reflectance Prior 2: It is more likely that the spectra will be similar for targets with similar heights. Thus, the reflectance of the pixel i can be expressed as

$$R_i = h_{ij}R_j \tag{19}$$

where R_i and R_j denote the reflectance of the ith and the jth pixels, respectively; and h_{ij} is the height similarity between the pixel i and the pixel j. The matrix form can be given as

$$\mathbf{R} = \mathbf{H}\mathbf{R}. \tag{20}$$

Considering the two constraint terms above, $\Psi(\mathbf{R})$ can be expressed as

$$\Psi(\mathbf{R}) = \theta_w\|\mathbf{R} - \mathbf{W}\mathbf{R}\|_F^2 + \theta_h\|\mathbf{R} - \mathbf{H}\mathbf{R}\|_F^2 \tag{21}$$

where θ_w and θ_h are weights of the two reflectance priors.

Abundance Prior: The most common abundance prior is the sparsity constraints of the abundance. The matrix form can be given as

$$\Upsilon(\mathbf{A}) = \gamma\|\mathbf{A}\|_{1,1} \tag{22}$$

where γ is a penalty parameter.

C. Optimization Algorithm

The basic objective function of the proposed method can be expressed as

$$\begin{aligned} U = &\frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\alpha}{2}\left\|\mathbf{S} - \mathbf{G}\mathbf{M}\mathbf{1}_N^T\right\|_F^2 + \frac{\beta}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 \\ &+ \theta_w\|\mathbf{R} - \mathbf{W}\mathbf{R}\|_F^2 + \theta_h\|\mathbf{R} - \mathbf{H}\mathbf{R}\|_F^2 + \gamma\|\mathbf{A}\|_{1,1} \end{aligned} \tag{23}$$

$$\text{s.t. } \mathbf{A} \geq 0$$

$$\arg\min_{\mathbf{S},\mathbf{R},\mathbf{A}} U(\mathbf{S}, \mathbf{R}, \mathbf{A}). \tag{24}$$

Noted for the component $\mathbf{S}$ composed of $\mathbf{G}$ and $\mathbf{M}$, $\mathbf{M}$ can be computed by the conjugate gradient method at the root of the position of the first partial derivatives [46]. This section details the optimization procedure used to solve (24) by the alternating direction method of multipliers (ADMM).

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First, by introducing auxiliary parameters $\mathbf{R} = \mathbf{J}$ and $\mathbf{A} = \mathbf{K}$, the solution of the objective variable can be obtained by calculating the minimum error. The reflectance component $\mathbf{R}$ is normally initiated to $\mathbf{Y}$

$$\begin{aligned}\min_{C} \frac{1}{2}\|\mathbf{Y}-\mathbf{S}\odot\mathbf{R}\|_F^2+\frac{1}{2}\|\mathbf{S}-\mathbf{GM}\|_F^2+\frac{1}{2}\|\mathbf{R}-\mathbf{EA}\|_F^2 \\ +\theta_w\|\mathbf{R}-\mathbf{WR}\|_F^2+\theta_h\|\mathbf{R}-\mathbf{HR}\|_F^2+\gamma\|\mathbf{K}\|_{1,1} \\ \text{s.t. } \mathbf{R}=\mathbf{J},\mathbf{A}=\mathbf{K},\mathbf{1}^T\mathbf{A}=\mathbf{1}\end{aligned} \tag{25}$$

where $C = \{\mathbf{R}, \mathbf{A}, \mathbf{J}, \mathbf{K}\}$.

Second, the augmented Lagrangian function can be expressed as

$$\begin{aligned}&L(C,\Lambda_1,\Lambda_2)\\ &=\frac{1}{2}\|\mathbf{Y}-\mathbf{S}\odot\mathbf{R}\|_F^2+\frac{1}{2}\|\mathbf{S}-\mathbf{GM}\|_F^2+\frac{1}{2}\|\mathbf{R}-\mathbf{EA}\|_F^2\\ &\quad+\theta_w\|\mathbf{R}-\mathbf{WR}\|_F^2+\theta_h\|\mathbf{R}-\mathbf{HR}\|_F^2+\gamma\|\mathbf{K}\|_{1,1}\\ &\quad+\Lambda_1^T(\mathbf{J}-\mathbf{R})+\Lambda_2^T(\mathbf{K}-\mathbf{A})\\ &\quad+\frac{\sigma_1}{2}\|\mathbf{J}-\mathbf{R}\|_F^2+\frac{\sigma_2}{2}\|\mathbf{K}-\mathbf{A}\|_F^2\\ &\text{s.t. } \mathbf{1}^T\mathbf{A}=\mathbf{1}\end{aligned} \tag{26}$$

where Λ_1 and Λ_2 are Lagrange multipliers, and σ_1 and σ_2 are positive penalty parameters.

Finally, the optimal solution of the variables involved is as follows.

Optimization with respect to $\mathbf{R}$:

$$\begin{aligned}\min_{\mathbf{R}} \frac{1}{2}\|\mathbf{Y}-\mathbf{S}\odot\mathbf{R}\|_F^2+\frac{1}{2}\|\mathbf{R}-\mathbf{EA}\|_F^2 \\ +\theta_w\|\mathbf{R}-\mathbf{WR}\|_F^2+\theta_h\|\mathbf{R}-\mathbf{HR}\|_F^2 \\ +\Lambda_1^T(\mathbf{J}-\mathbf{R})+\frac{\sigma_1}{2}\|\mathbf{J}-\mathbf{R}\|_F^2\end{aligned} \tag{27}$$

which can also be expressed as

$$\begin{aligned}\min_{\mathbf{R}} \frac{1}{2}\|\mathbf{T}-\mathbf{R}\|_F^2+\frac{1}{2}\|\mathbf{R}-\mathbf{EA}\|_F^2 \\ +\theta_w\|\mathbf{R}-\mathbf{WR}\|_F^2+\theta_h\|\mathbf{R}-\mathbf{HR}\|_F^2 \\ +\Lambda_1^T(\mathbf{J}-\mathbf{R})+\frac{\sigma_1}{2}\|\mathbf{J}-\mathbf{R}\|_F^2\end{aligned} \tag{28}$$

where $\mathbf{S}\odot\mathbf{T} = \mathbf{Y}$, $\|\mathbf{Y}-\mathbf{S}\odot\mathbf{R}\|_F^2$ can be equivalently described as $\|\mathbf{T}-\mathbf{R}\|_F^2$.

Then, $\mathbf{R}$ can be computed by

$$\begin{aligned}\mathbf{R}=\frac{1}{2(\sigma_1+1)}(\mathbf{EA}+\mathbf{T}-\mathbf{T}_0\mathbf{T}_1+\mathbf{T}_0\mathbf{W}^T\mathbf{T}_1-\mathbf{T}_2\mathbf{T}_3 \\ +\mathbf{T}_2\mathbf{H}^T\mathbf{T}_3+2\sigma_1\mathbf{J}+\Lambda_1\mathrm{I}).\end{aligned} \tag{29}$$

Here, four variables have been used to optimize the expression

$$\begin{aligned}\mathbf{T}_0&=2\mathbf{W}\\ \mathbf{T}_1&=\mathbf{R}-\mathbf{WR}\\ \mathbf{T}_2&=2\mathbf{H}\\ \mathbf{T}_3&=\mathbf{R}-\mathbf{HR}.\end{aligned} \tag{30}$$

Optimization concerning $\mathbf{A}$

$$\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{R}-\mathbf{EA}\|_F^2+\Lambda_2^T(\mathbf{K}-\mathbf{A})+\frac{\sigma_2}{2}\|\mathbf{K}-\mathbf{A}\|_F^2 \tag{31}$$

which are well addressed by

$$\mathbf{A}=(2\sigma_2\mathbf{K}-\mathbf{E}^T\mathbf{R}-\Lambda_2\mathbf{I})\times(2\sigma_2-\mathbf{E}^T\mathbf{E})^{-1}. \tag{32}$$

Optimization for $\mathbf{J}$:

$$\min_{\mathbf{J}} \Lambda_1^T(\mathbf{J}-\mathbf{R})+\frac{\sigma_1}{2}\|\mathbf{J}-\mathbf{R}\|_F^2 \tag{33}$$

with analytical resolution

$$\mathbf{J}=\sigma_1\mathbf{R}-\Lambda_1/\sigma_1. \tag{34}$$

Optimization to $\mathbf{K}$:

$$\min_{\mathbf{K}}\|\mathbf{K}\|_1+\Lambda_2^T(\mathbf{K}-\mathbf{A})+\frac{\sigma_2}{2}\|\mathbf{K}-\mathbf{A}\|_F^2 \tag{35}$$

which can be inferred by employing a soft-threshold operator as

$$\mathbf{K}=\mathrm{sign}(\mathbf{A}-\Lambda_2/\sigma_2)\odot\max\{\mathbf{0},|\mathbf{A}-\Lambda_2/\sigma_2|-1/\sigma_2\}. \tag{36}$$

Optimization with respect to Λ_2 and Λ_2:

The Lagrange multipliers Λ_1 and Λ_2 can be updated as follows:

$$\begin{aligned}\Lambda_1&\leftarrow\Lambda_1+\sigma_1(\mathbf{J}-\mathbf{R})\\ \Lambda_2&\leftarrow\Lambda_2+\sigma_2(\mathbf{K}-\mathbf{A}).\end{aligned} \tag{37}$$

IV. Experimental Results and Discussion

A. Data Description

To validate the performance of the proposed DSM data-embedded intrinsic hyperspectral unmixing algorithm, three HSI datasets, i.e., the pseudo-synthetic dataset, the Houston dataset, as well as the MUFFLE dataset are used.

1) *Pseudo-Synthetic Dataset [46]:* The pseudo-synthetic dataset has been applied to quantitatively evaluate the performance of IID, which was acquired from the "grss_dfc_2018" dataset used for the 2018 IEEE GRSS Data Fusion Contest. The dataset contains three endmembers: *#1 Road*, *#2 Tree*, and *#3 Shadow*. Fig. 2(a) presents a false color image of this dataset and the DSM data.
2) *Houston Dataset [23]:* An original 349 × 1905 pixel HSI was acquired by the ITRES CASI-1500 sensor over the University of Houston campus, USA, in June 2012. The ITRES CASI-1500 sensor contains 144 bands ranging from 0.4 to 1.0 μm. A 170 × 170 pixel subimage was selected from the original image. The four endmembers included in this multimodal dataset are *#1 Parking lot1*, *#2 Parking lot2*, *#3 Running track*, and *#4 Healthy Grass*. Fig. 2(b) presents this multimodal hyperspectral dataset.
3) *MUFFLE Dataset [23]:* An original 325 × 220 pixel HSI was collected over the campus of Southern Mississippi Gulfpark [54], and 64 spectral bands with a spectral coverage ranging from 0.4 to 1.0 μm were captured. A 130 × 90 pixel region of interest (ROI) was selected from the original image. Five endmembers considered in the region are *#1 Roof*, *#2 Grass*, *#3 Tree*, *#4 Shadow*, and *#5 Asphalt*. Fig. 2(c) presents a false color image of this dataset and the corresponding DSM data.

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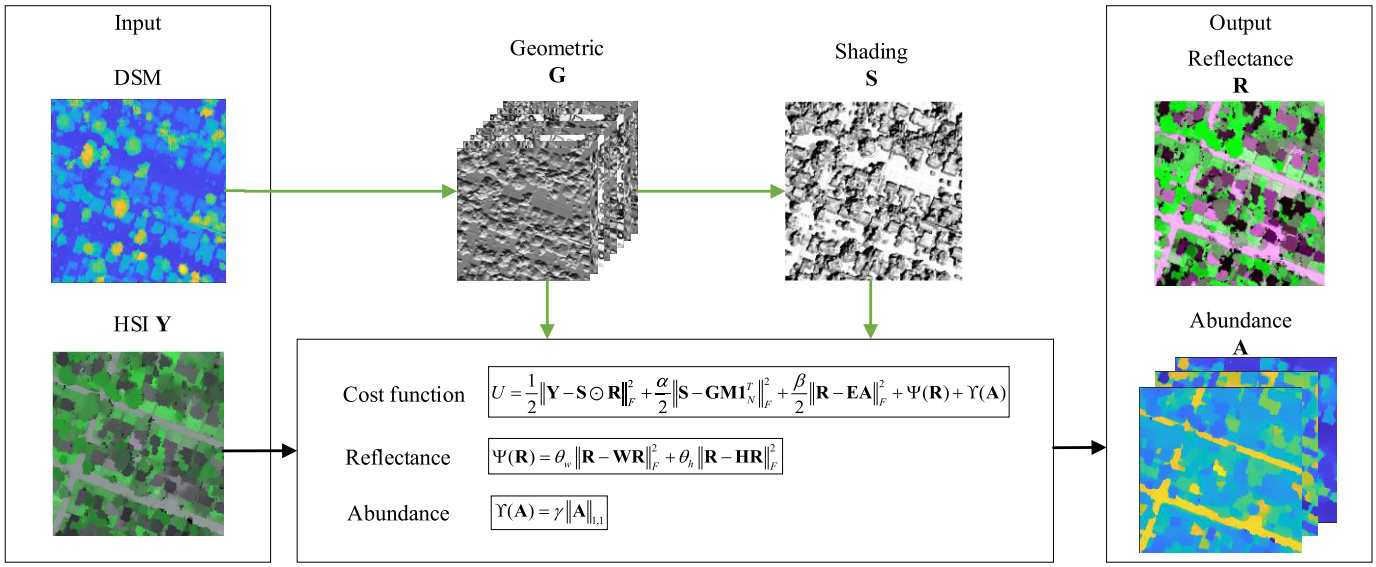


Fig. 1. Framework of the proposed DSM data-embedded intrinsic hyperspectral unmixing algorithm. The geometric component can be computed by the DSM data; then, the shading component can be recovered, then by imposing priori constraint terms on the reflectance component and abundance component; the reflectance component and the abundance component can be estimated.

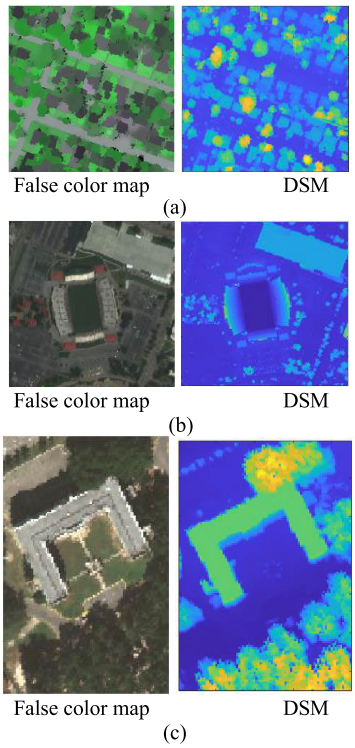


Fig. 2. False color maps and DSM data for the three datasets. (a) Pseudo-synthetic dataset. (b) Houston dataset. (c) MUFFLE dataset.

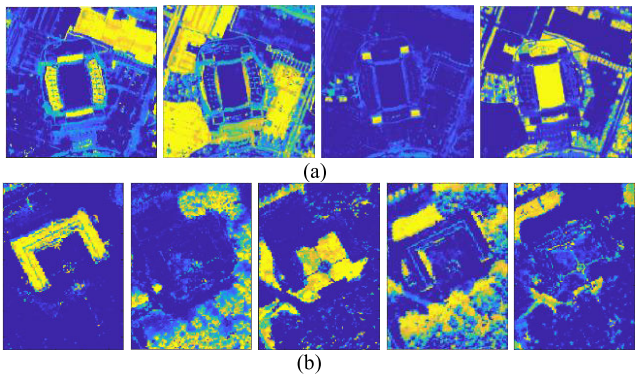


Fig. 3. Abundance map for the two datasets. (a) Houston dataset, four endmembers considered in the region are #1 Parking lot1, #2 Parking lot2, #3 Running track, and #4 Healthy Grass. (b) MUFFLE dataset, five endmembers considered in the region are #1 Roof, #2 Grass, #3 Tree, #4 Shadow, and #5 Asphalt.

B. Experimental Setting

To verify the effectiveness of the proposed algorithms in this article, seven typical unmixing methods have been incorporated into the proposed method from the following four considerations.

1) *Model-Driven Linear Unmixing Method for Hyperspectral Data:* Fully constraint least-squares (FCLS), Sparse SU [14], and extended LMM (ELMM) [30]. FCLS is a baseline unmixing algorithm where the ASC and ANC constraints are included. The abundance estimated by FCLSU is generally used as initialization for other algorithms. The Sparse SU method uses a prior spectral library to perform the linear unmixing algorithms. The number of endmembers in the prior spectral library is larger than those in the image, there is no need to extract or estimate the number of endmembers in the image. ELMM is a typical unmixing method under spectral variability using a scaling factor for variable spectra.
2) *Model-Driven Nonlinear Unmixing Method for Hyperspectral Data:* The intrinsic image decomposition-embedded spectral unmixing (IIDSU) [32] and the robust unmixing by variable splitting and augmented Lagrangian (RUSAL) [33]. The IIDSU aims to address the SU problem under the phenomenon of spectral variability in single-modality HSIs. The RUSAL supplements the component that cannot be fit by the LMM with additional nonlinear terms.

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TABLE I
RECONSTRUCTION RESULTS OF DIFFERENT METHODS FOR THE THREE DATASETS

Datasets		FCLSU	Sparse	RUSAL	ELMM	IIDSU	uDAS	MUNET	Proposed
pseudo synthetic dataset	PSNR	12.31	22.09	20.04	26.08	19.86	36.99	37.63	**38.69**
	RMSE	7.32e-05	2.96e-05	3.26e-05	2.16e-05	3.92e-05	9.60e-06	6.96e-06	**3.81e-06**
Houston dataset	PSNR	25.35	25.95	35.7	28.61	27.68	23.19	34.97	**36.39**
	RMSE	1.93e-05	1.85e-05	6.09e-06	1.34e-05	1.32e-05	3.45e-05	9.14e-06	**5.22e-06**
MUFFLE dataset	PSNR	30.71	39.63	38.36	31.95	37.93	37.98	22.68	**47.09**
	RMSE	3.07e-05	1.17e-05	1.45e-05	3.07e-05	4.38e-05	1.79e-05	8.45e-05	**5.69e-06**

3) *Data-Driven Unmixing Method for Hyperspectral Data:* The uDAS method [20] incorporates nonnegative sparsity and denoising coding modules to reduce noise interference.
4) *Data-Driven Unmixing Method for Multimodal Hyperspectral Data:* Multimodal unmixing network (MUNET) [23], by incorporating the SE attention into the AE unmixing network, can fuse HSI and LiDAR features effectively.

It is worth mentioning that the parameter settings for the above comparative approaches refer to the original literature in this article, and the initial endmembers are all extracted by VCA [55] for fair comparison. θ_w and θ_h are set to 50 and 10, respectively [42]. To facilitate the evaluation of abundance root-mean-square error (aRMSE), all comparison methods will eventually meet the ASC.

C. Evaluation Criteria

In this article, the evaluation of the different SU algorithms is considered from two main aspects.

First, from the reconstruction point of view, the reconstruction overall root-mean-square error (RMSE) and the peak signal-to-noise ratio (PSNR) are introduced for evaluating the magnitude of the error between the product of endmember and abundance generated by the unmixing and the original HSI. RMSE is considered to be the smaller, the better, while PSNR is considered to be the larger, the better.

Second, from the unmixing point of view, the aRMSE is considered. aRMSE is regarded as small as possible.

The evaluation criteria are calculated using the following formula:

$$\mathrm{RMSE}(Y_i, Y_j) = \sqrt{\frac{1}{N}\sum \left\|Y_i - Y_j\right\|_2} \tag{38}$$

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{MAX^2}{\mathrm{MSE}}\right) \tag{39}$$

where

$$\mathrm{MSE} = \frac{1}{nn} \cdot \sum_{i=0}^{n-1}\sum_{j=0}^{n-1}(Y_i - Y_j)^2 \tag{40}$$

$$\mathrm{aRMSE}(a_i, a_j) = \sqrt{\frac{1}{N}\sum \left\|a_i - a_j\right\|_2}. \tag{41}$$

The pseudo-synthetic dataset has no reference abundance values, while the Houston dataset and the MUFFLE dataset have referable values of abundance, as shown in Fig. 3. Thus, the pseudo-synthetic dataset is evaluated with respect to reconstruction, while the Houston dataset and the MUFFLE dataset are evaluated with respect to both reconstruction and unmixing.

D. Experimental Results and Analysis

First, the experimental results can be analyzed qualitatively from the visualization. Fig. 4 shows the visual reconstruction performance of three datasets by different unmixing algorithms. Figs. 5–7 display the visual abundance maps of the three datasets by different unmixing algorithms. It can be seen from Fig. 5 that shadow can be effectively removed, and in addition, the abundance of trees and roads is clearly distinguishable and the spatial geometry is well maintained. In Fig. 6, the model-driven linear unmixing methods and the model-driven nonlinear unmixing method perform poorly on abundance estimation of Parking lot1, while the multimodal unmixing methods can get better results. In Fig. 7, the ELMM and the multimodal unmixing methods can achieve better abundance estimation of the roof and tree. The data-driven unmixing method uDAS performs better on pseudo-synthetic data and meanwhile performs poorly on real data, which can be attributed to the fact that the noise distribution of the real dataset is more complex and the denoising module has difficulty in modeling the noise accurately. The abundance estimation of the MUNET method can all achieve relatively good experimental results; however, it is a pity that the reconstruction results are normally effective and a lot of detailed information is smoothed out, leading to the loss of information.

Second, it is possible to quantitatively analyze the experimental results of the evaluation criteria. Table I details the results of the quantitative reconstruction results for each of the three datasets in terms of reconstruction and Table II details the results of the quantitative unmixing results for the Houston dataset and the MUFFLE dataset. The best experimental results are shown in bold. As can be seen from Table I, the reconstruction result of [24] is general, which shows the same direction as the results of the visualization. As for the pseudo-synthetic dataset, since there is no corresponding referenceable abundance, the results are based on the qualitative reconstruction error and abundance maps, as well as quantitative metrics such as reconstruction error and PSNR in Table I to assess the quality of results. Due to the low original signal-to-noise ratio of the pseudo-synthetic dataset, it is difficult to preserve geometric shapes effectively during the unmixing process. For the Houston dataset and the MUFFLE real dataset, the physics-based nonlinear SU method

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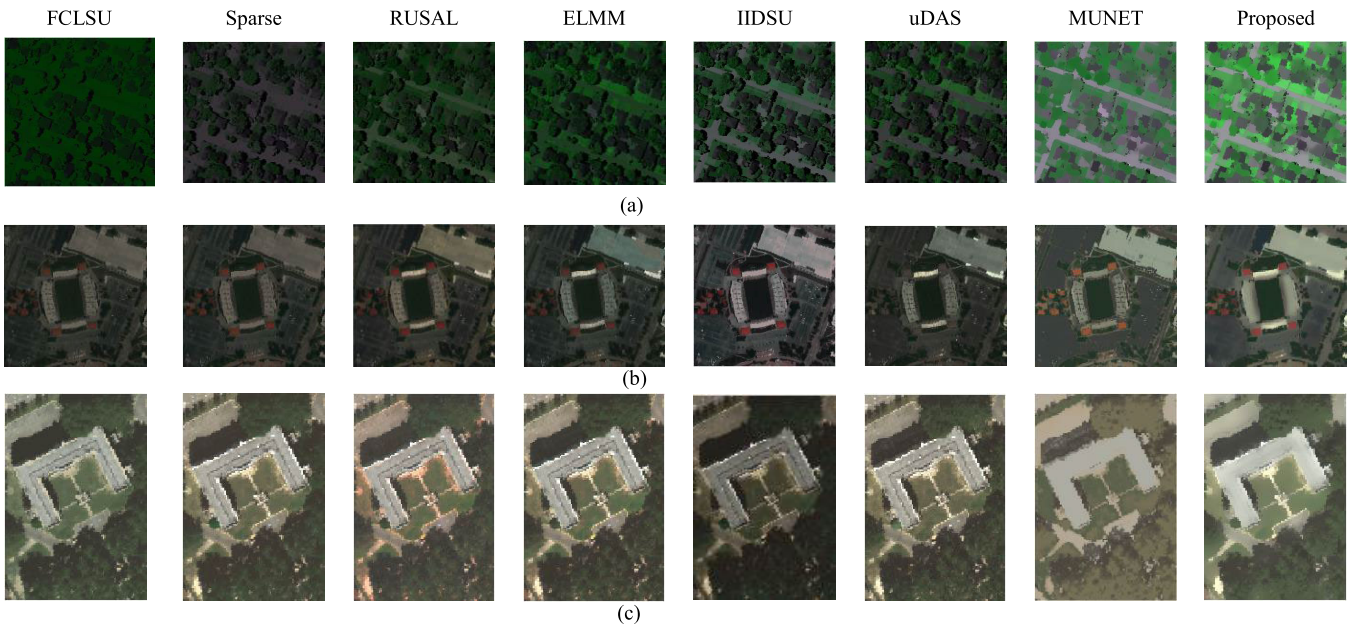


Fig. 4. Reconstructed results of the three datasets obtained by different algorithms. (a) Pseudo-synthetic dataset. (b) Houston dataset. (c) MUFFLE dataset.

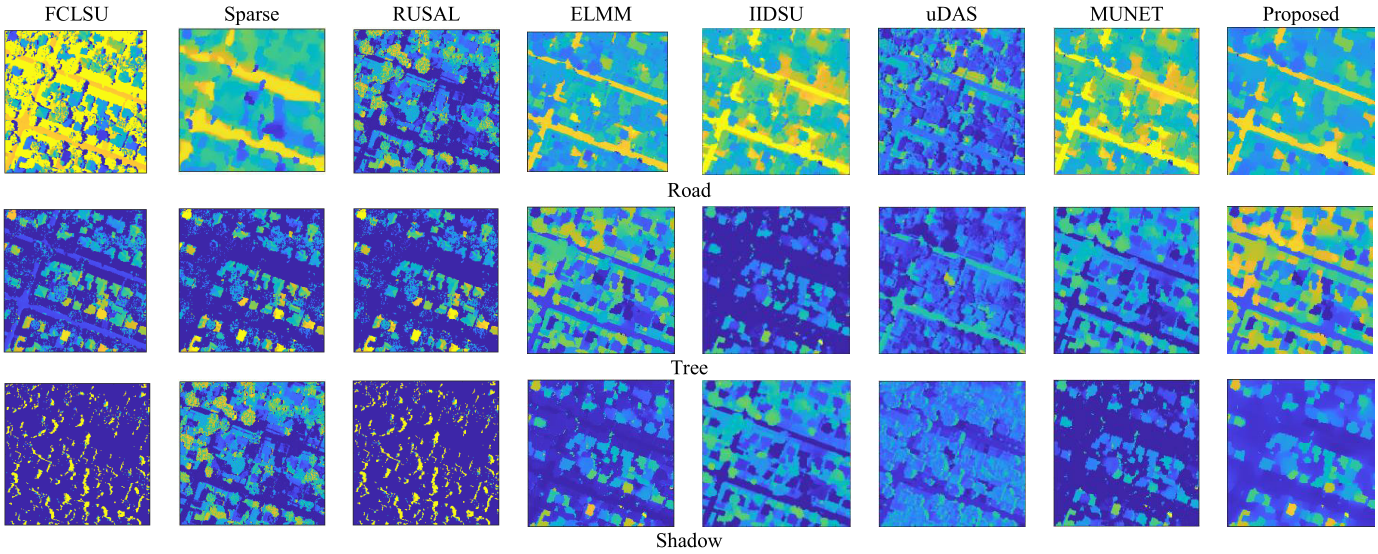


Fig. 5. Abundance maps of pseudo-synthetic dataset (three materials) obtained by different algorithms.

RUSAL [35] achieves promising experimental results, whereas the deep learning-based multimodal SU method [23] does not yield satisfactory outcomes, possibly due to loss of detailed information during processing. For the three datasets used, the proposed method demonstrates superior performance in reconstruction. As can be seen from Table II, the proposed method can obtain minimum aRMSE. It is also possible to obtain accurate experimental results for the data-driven multimodal method [24].

Third, the proposed model consists of two main components. The imaging model term and the prior constraint terms, which specifically include the DSM term (D term), the local reflectance constraint (R term), and the global abundance constraint (A term). To validate the effectiveness of each constraint within the proposed model, six ablation experiments were conducted across the three datasets. Given the physical significance of the imaging model term and the constraints, the ablation experiments were set up as follows: ablation of the D term, ablation of the R term, ablation of the A

TABLE II
aRMSE OF DIFFERENT ALGORITHMS FOR THE THREE DATASETS

Methods	Houston dataset	MUFFLE dataset
FCLSU	0.0049	0.1043
Sparse	0.0040	0.0099
RUSAL	0.0023	0.0153
ELMM	0.0096	0.0099
IIDSU	0.0083	4.0584
uDAS	0.0524	0.0242
MUNET	0.0016	0.0033
Proposed	**0.0014**	**0.0023**

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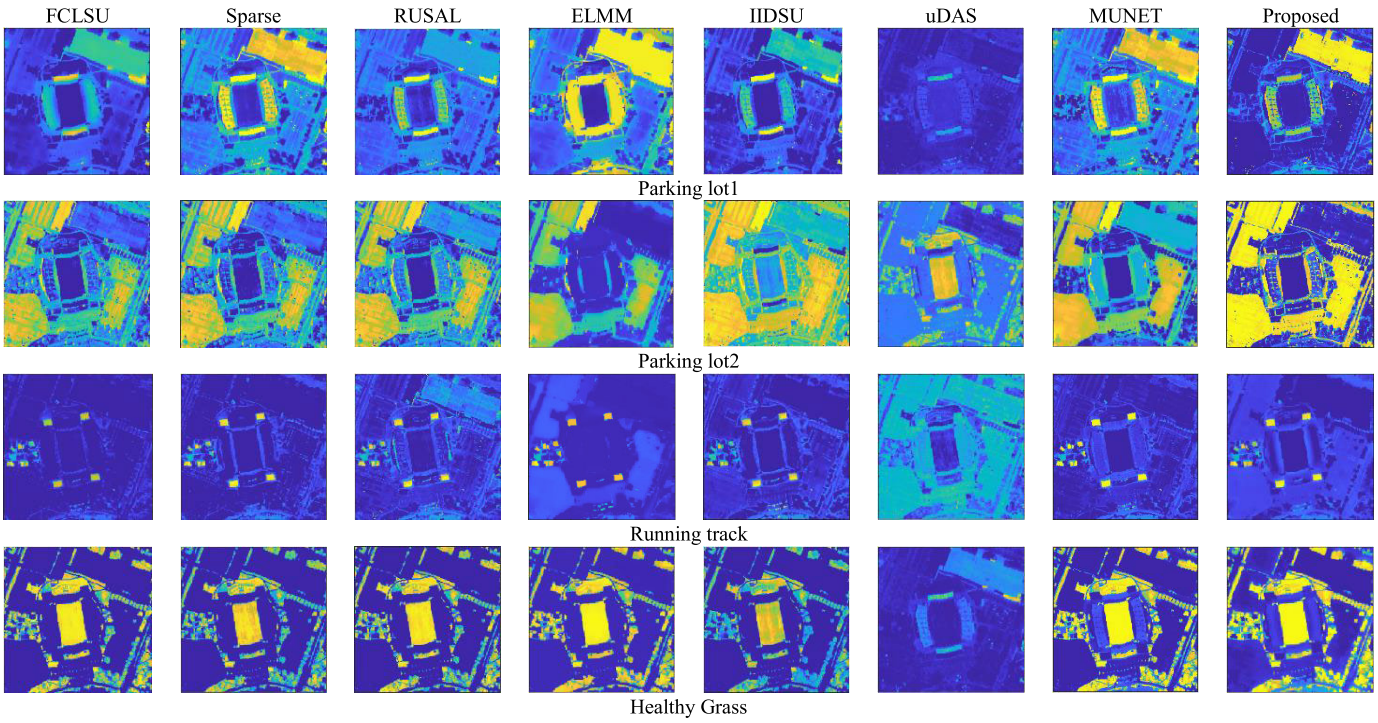


Fig. 6. Abundance maps of Houston dataset (four materials) obtained by different algorithms.

term, and ablation of both the R and A terms. Table III presents the RMSE values obtained from these different ablation experiments. The "Baseline" model serves as the reference framework, which does not include any additional modules. The "Baseline + D" model represents the baseline framework with the addition of DSM data. As shown in Table III, the RMSE value of the "Baseline" model is the highest. Compared to the "Baseline" model, the "Baseline + DSM," "Baseline + R," and "Baseline + A" models show significant reductions in RMSE values. These results validate the effectiveness of the three modules, with DSM data assistance and global abundance constraints contributing the most to model improvement. The table further indicates that the joint use of these modules enhances the performance of SU. In summary, the method proposed in this article integrates DSM data into the imaging model term and effectively incorporates the constraint terms into a unified framework, achieving optimal SU results.

The most time-consuming process of the proposed method is to optimize the cost function, which is also related to the number of hyperspectral pixels. The computational complexity is O(n). In addition, the computational performance is shown in Table IV, which is evaluated in seconds. As can be seen from Table IV, FCLSU is the fastest method because there are no iterations in the experimental process. Among all other methods of comparison, the proposed method does not achieve the best computation time, but its computing time is perfectly acceptable compared to other comparative methods. The convergence of the proposed method for the three datasets is shown in Fig. 8. As we can see from Fig. 8, all three datasets exhibit the characteristic of residuals rapidly converging and eventually stabilizing as the number of iterations increases during the optimization process. Based on the convergence plot of residuals, it can be easily inferred that a small number of iterations can still yield satisfactory experimental results, which verifies the better convergence of the proposed method.

In addition, Fig. 9 presents a sensitivity analysis of the proposed method, specifically examining the effect of the regularization parameter γ on performance with the MUFFLE dataset. Through a series of systematic experiments, we observed that the RMSE error achieves its minimum

TABLE III
RMSE OF ABLATION EXPERIMENTS ON THE THREE DATASETS

METHODS/RMSE	synthetic dataset	Houston dataset	MUFFLE dataset
BASELINE	7.32e-05	4.03e-05	8.96e-05
BASELINE+D	5.30e-05	1.85e-05	3.70e-05
BASELINE+R	6.76e-05	1.93e-05	4.38e-05
BASELINE+A	3.70e-05	1.49e-05	3.07e-05
BASELINE+R+A	2.96e-05	1.32e-05	1.17e-05
BASELINE+D+R+A	3.81e-06	5.22e-06	5.69e-06

TABLE IV
COMPUTING TIME OF DIFFERENT ALGORITHMS FOR THE THREE DATASETS

Methods	synthetic dataset	Houston dataset	MUFFLE dataset
FCLSU	4.79	3.92	1.82
Sparse	45.63	59.11	23.61
RUSAL	69.75	30.79	22.13
ELMM	232.9	569.80	143.92
IIDSU	45.19	54.06	21.67
uDAS	65.78	226.28	18.61
MUNET	68.10	34.97	52.86
Proposed	76.76	90.94	43.96

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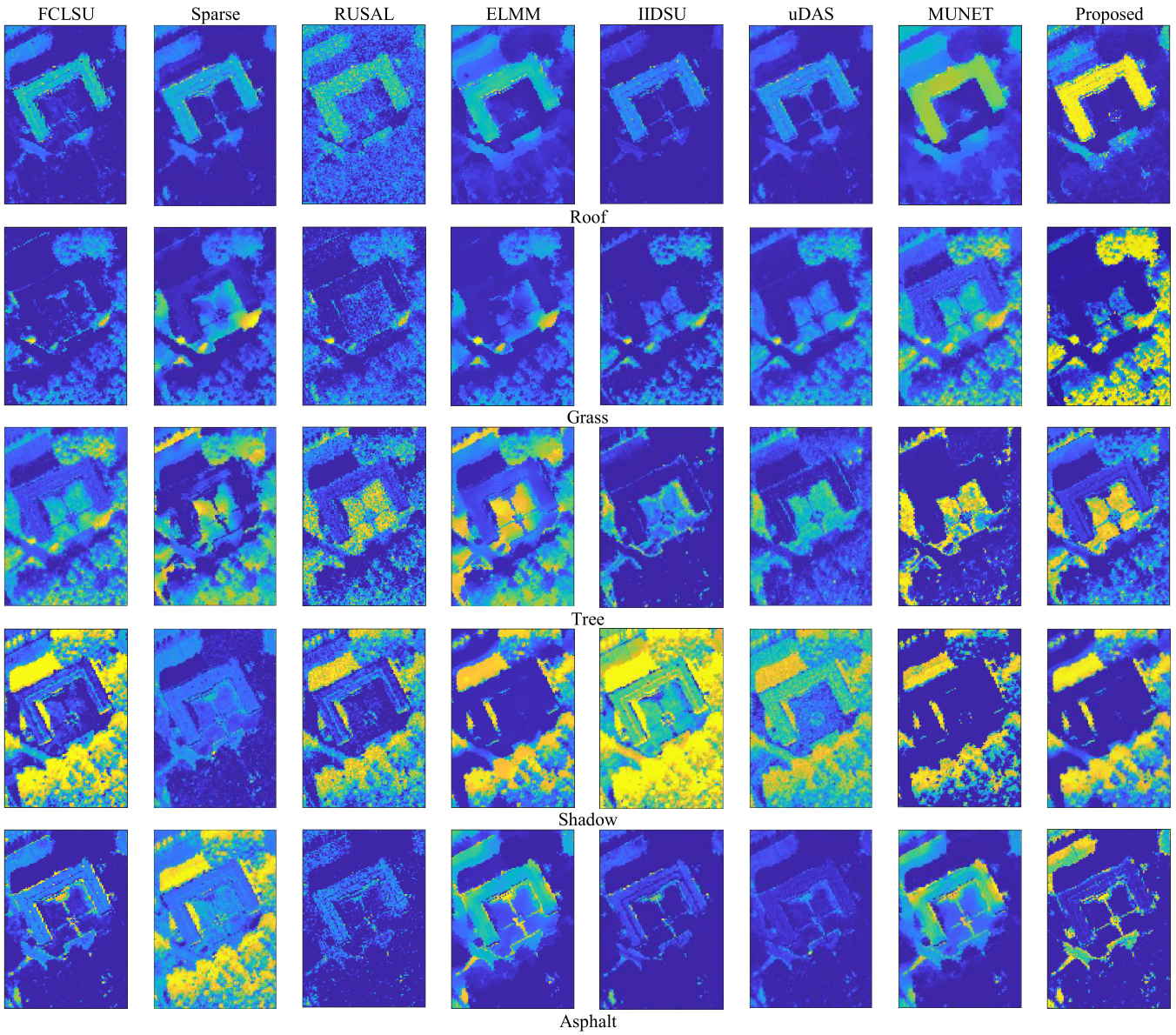


Fig. 7. Abundance maps of MUFFLE dataset (five materials) obtained by different algorithms.

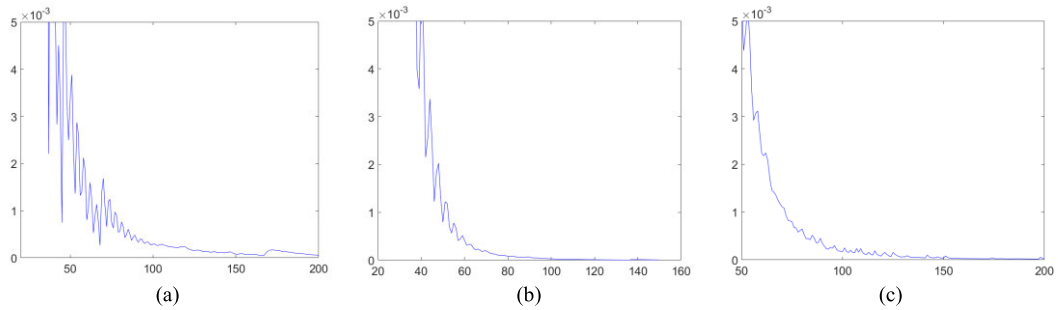


Fig. 8. Convergence plot of residuals for the three datasets. (a)–(c) Overall trend of the pseudo-synthetic dataset, the Houston dataset, and the MUFFLE dataset, respectively. The vertical axis represents residuals, while the horizontal axis denotes the number of iterations.

value when γ is set to 10^{-3}, demonstrating the optimal balance between model complexity and regularization strength.

Since the comparison methods represent different perspectives and show the effectiveness of the different datasets, it can be concluded that the proposed method can obtain more accurate abundance estimation results as well as the most superior reconstruction performance, which demonstrates that the method proposed can efficiently achieve optimal multimodal SU results in real-world scenarios.

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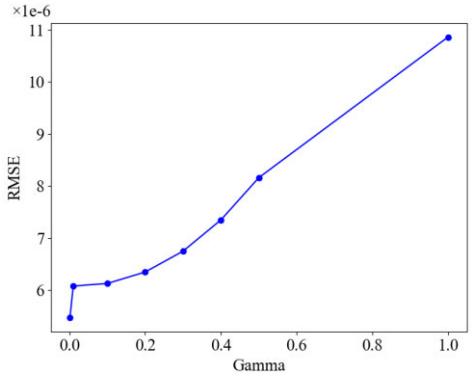


Fig. 9. Parameter sensitivity analysis for λ of the proposed method for the MUFFLE dataset.

V. Conclusion

In this article, a novel DSM-embedded intrinsic hyperspectral unmixing method is proposed to incorporate elevation information of the DSM data into the intrinsic unmixing model from a physical imaging perspective. In this way, the shading image can be estimated more realistically. By introducing prior information on reflectance information and priori information on the abundance, the reflectance component can be recovered more accurately; thus, the effects caused by spectral variability can be effectively attenuated, and the results of abundance estimation will be more precise. The qualitative and quantitative experimental results on three multimodal datasets can demonstrate the effectiveness and superiority of the proposed method. In addition, from the physical modeling point of view, the hyperspectral unmixing with the embedded DSM data offers great possibilities for the multimodal unmixing task. In future work, we intend to explore the potential of incorporating intensity data from point cloud datasets to inform the SU process, with the objective of obtaining subpixel level abundance information.

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